Superior High-Temperature Electrical Characteristics of ALD Ultrathin In_2O_3 Transistors

Tingting Gao, Peiyan Hong, Kaisheng Hu, Xiangshui Miao, *Senior Member, IEEE*, Yuhui He, *Member, IEEE*, and Xuefei Li

***Abstract*— In this work, we report an effective method to improve the high-temperature reliability of atomic-layer-deposited (ALD) ultrathin (3 nm) indium oxide (In_2O_3) transistors. A high field-effect mobility (μ_{FE}) of 80 $cm^2/V{\cdot}s$ at 85 °C and an ultralow average rate of voltage shift of −0.8 mV/K were successfully demonstrated, which was attributed mainly to the efficient passivation of oxygen vacancies in the In_2O_3 channel by O_3-based atomic layer deposition of HfO_2, ultrathin channels, and high capacitance density. Excellent bias-thermal stability characteristics of In_2O_3 transistors were achieved. Low-frequency (1/*f*) noise is generated mainly in the channel region, and the noise level remains almost the same from 25 °C to 85 °C, indicating excellent interface quality and low trap density.**

***Index Terms*— Indium oxide, HfO_2 passivation, reliability, low frequency noise.**

I. Introduction

AMORPHOUS oxide semiconductors are compatible with back-end-of-line (BEOL) processing [1] and monolithic three-dimensional (3D) integration because of their excellent large area uniformity and low-temperature deposition [2], [3], [4], [5]. Moreover, these materials are promising candidates as channel materials in field-effect transistors (FETs) because of their high carrier mobilities and low off-state currents [2], [3], [6], [7]. High-performance back-gate In_2O_3 transistors with $\sim$ 3 mA/μm at a channel length (L_{ch}) of 8 nm have been demonstrated [8]. On the other hand, the effects of temperature on the electrical characteristics of oxide semiconductor devices have been reported by many groups [9], [10], [11], [12], [13], [14]. A high temperature may lead to a shift in the threshold voltage (V_{th}), mobility, and subthreshold swing.

Received 8 September 2024; revised 24 September 2024; accepted 27 September 2024. Date of publication 1 October 2024; date of current version 26 November 2024. This work was supported by the National Key Research and Development Program of China under Grant 2023YFB4502200. The review of this letter was arranged by Editor S. Zhang. *(Corresponding authors: Yuhui He; Xuefei Li.)*

Tingting Gao, Xiangshui Miao, and Yuhui He are with the School of Integrated Circuits, Huazhong University of Science and Technology, Wuhan 430074, China (e-mail: heyuhui@hust.edu.cn).

Peiyan Hong, Kaisheng Hu, and Xuefei Li are with Wuhan National High Magnetic Field Center and the School of Integrated Circuits, Huazhong University of Science and Technology, Wuhan 430074, China (e-mail: xfli@hust.edu.cn).

Color versions of one or more figures in this letter are available at https://doi.org/10.1109/LED.2024.3471634.

Digital Object Identifier 10.1109/LED.2024.3471634

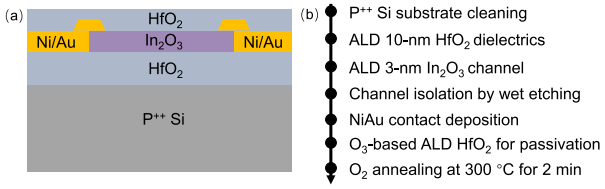


Fig. 1. (a) Cross-sectional schematic view of the back-gate In_2O_3 transistor with a passivation layer. (b) Fabrication process of back-gate In_2O_3 transistors with a passivation layer.

Recently, the temperature-dependent I-V characteristics of In_2O_3 transistors have also been reported to exhibit an obvious threshold voltage shift in the range of 20 $\sim$ 100 °C [8], [15]. However, an effective method for passivation and improving the high-temperature stability of ultrathin ALD In_2O_3 transistors remains to be explored.

In this work, we fabricate back-gate In_2O_3 transistors with a high-quality HfO_2 passivation layer, which is deposited via O_3-based ALD. The temperature-dependent I-V characteristics are measured for all channel lengths in the range of 25 $\sim$ 85 °C. An ultralow average rate of voltage shift of −0.8 mV/K for the 0.2 μm channel length device is achieved, and the activation energy is 0.27 eV. Furthermore, low-frequency noise (LFN) measurements and negative bias stress (NBS) and positive bias stress (PBS) measurements are performed to investigate the interface quality and bias-thermal stability at high temperatures.

II. Device Fabrication

A schematic of the fabricated In_2O_3 transistors is shown in Fig. 1(a). The channel length (L_{ch}) of the In_2O_3 transistors ranges from 0.2 to 2 μm, and the channel width (W_{ch}) is fixed at 0.6 μm. The fabrication process of the In_2O_3 transistor is schematically shown in Fig. 1(b). First, the back-gate dielectric is formed by ALD 10 nm HfO_2 on a p^{++} silicon substrate. A 3-nm-thick In_2O_3 film was then deposited by ALD at 225 °C by controlling the cycle number using $(CH_3)_3$ In (TMIn) and H_2O as the In and O precursors, respectively. The channel region was defined via electron-beam lithography (EBL), followed by channel isolation via wet etching with concentrated HCl. The 10/20 nm Ni/Au was then deposited by e-beam evaporation to form contacts. A previous study has shown that O_3 is more useful for achieving a positive V_{th} [16]. To balance the trade-off between mobility and V_{th}, 250 °C was chosen to deposit 4-nm HfO_2 passivation layers by ALD

0741-3106 © 2024 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Huazhong University of Science and Technology. Downloaded on February 04,2025 at 06:05:16 UTC from IEEE Xplore. Restrictions apply.

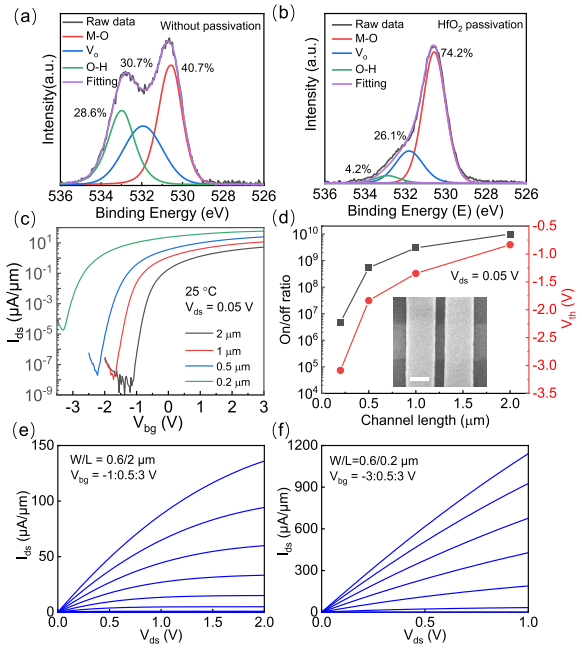


Fig. 2. O 1 s spectra of the devices (a) before and (b) after HfO_2 passivation. (c) Transfer characteristics of the In_2O_3 FET with different L_{ch} values at a V_{ds} of 0.05 V. (d) The function of the on/off ratio and V_{th} against the channel length. Inset: scanning electron microscope image of back-gate transistors with an L_{ch} of 200 nm. The scale bar is 400 nm. Output characteristics of the In_2O_3 transistor with L_{ch} values of (e) 2 μm, and (f) 0.2 μm at 25 °C.

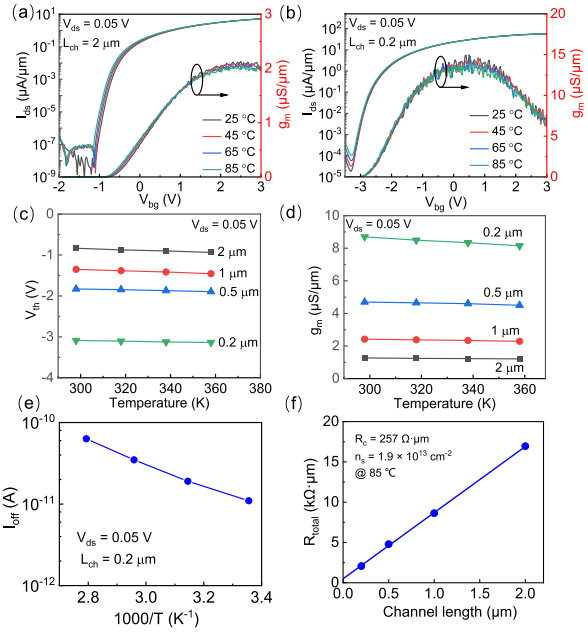


Fig. 3. Evolution of the transfer characteristics and transconductance of the device with varying temperature at channel lengths of (a) 2 μm, and (b) 0.2 μm. (c) Effect of temperature on V_{th} for different channel lengths. (d) Effect of temperature on g_m for different channel lengths. (e) Arrhenius plot of I_{off} as a function of temperature for an L_{ch} of 0.2 μm. (f) The extracted R_c of In_2O_3 transistors with a HfO_2 passivation layer at $n_s = 1.9 \times 10^{13}$ cm^{-2}.

using $[CH_3CH_2(CH_3)N]_4Hf$ and O_3 as Hf and O precursors. Finally, the fabricated devices were annealed in O_2 for 2 min at 300 °C. The electrical characteristics of the In_2O_3 transistors were measured using a Lakeshore vacuum probe station with an Agilent semiconductor parameter analyzer B1500A. The $1/f$ noise power spectral density (S_{id}) of the In_2O_3 transistor was measured at a low V_{ds} of 0.05 V by an Agilent E4725A $1/f$ noise system with an Agilent E5052B.

III. Results and Discussion

To understand the physical mechanism, we analyzed the XPS spectra of both devices as shown in Fig. 2(a) and (b). The O 1 s spectrum shows three different peaks that are deconvoluted with Gaussian distributions, which can be attributed to fully oxidized oxygen (M-O), an oxygen vacancy (V_o), and a hydroxide state (O-H) [17]. This shows that O_3-based ALD HfO_2 can reduce the number of donor traps such as oxygen vacancies and hydrogen states. Fig. 2(c) shows the transfer characteristics of the fabricated In_2O_3 transistors with different channel lengths and V_{ds} = 0.05 V at room temperature. The V_{th} is extracted using the constant drain current method at V_{bg} where I_{ds} = 100 pA $\times W_{ch}/L_{ch}$. The transfer curves shift negatively as L_{ch} decreases due to the short-channel effect and high carrier density in the channel [4]. V_{th} can be further tuned by changing the annealing temperature and time as well as the pulse time of O_3 [18]. Fig. 2(d) shows the effect of the channel length on the on/off ratio and V_{th}. The on/off ratio of the device with a 2 μm channel length approaches 10^{10}, and the corresponding V_{th} is −0.83 V. As expected, when L_{ch} decreases to 0.2 μm, the on/off ratio decreases to $\sim 5 \times 10^7$ and V_{th} shifts negatively to −3 V. Fig. 2(e) and (f) display the output characteristics of the In_2O_3 transistor with L_{ch} values of 2 and 0.2 μm at 25 °C, respectively. A maximum I_{ds} of 1146 μA/μm is achieved at a V_{ds} of 1 V for the short-channel device.

Fig. 3(a) and (b) show the evolution of the transfer characteristics of the In_2O_3 device at different temperatures with channel lengths of 2 μm and 0.2 μm, respectively. The field-effect mobility was extracted from the maximum transconductance at low V_{ds} according to $\mu_{FE} = L_{ch} \cdot g_m (W_{ch} \cdot C_{ox} \cdot V_{ds})^{-1}$, where C_{ox} and g_m are the gate capacitance per unit area and the transconductance, respectively. The field-effect mobility (μ_{FE}) of the In_2O_3 transistor at 85 °C is $\sim$80 $cm^2/V \cdot s$. In addition, the V_{th} of the 0.2 μm channel length remains the same as the temperature increases, indicating excellent stability at high temperatures. Fig. 3(c) summarizes the effect of high temperature on V_{th} for different channel lengths. The average rate of shift, $\partial V_{th}/\partial T$, is approximately −0.8 mV/K for an L_{ch} of 0.2 μm, which is 67 times lower than the value of −53.9 mV/K reported for back-gate In_2O_3 devices without passivation [15]. Fig. 3(d) also shows the effect of high temperature on g_m for different channel lengths, indicating a weak temperature dependence. An Arrhenius plot of I_{off} as a function of temperature for an L_{ch} of 0.2 μm is shown in Fig. 3(e). The activation energy (E_a) can be extracted from the slope of the curves by the formula $I_{off} \propto \exp(-E_a/k_B T)$. In this way, E_a is calculated to be $\sim$0.27 eV, which is only a quarter of the value of the previously reported back-gate In_2O_3 transistor. This value is less than a tenth of the bandgap, indicating the excellent high-temperature stability of the In_2O_3 transistor after HfO_2 passivation. This might be attributed to the decrease in the subgap density-of-state (DOS) distribution caused by the passivation of oxygen vacancies, ultrathin In_2O_3 channels, and high dielectric capacitance density [19], [20]. We further calculated the contact resistance R_c

Authorized licensed use limited to: Huazhong University of Science and Technology. Downloaded on February 04,2025 at 06:05:16 UTC from IEEE Xplore. Restrictions apply.

TABLE I
EFFECT OF TEMPERATURES ON ELECTRICAL PROPERTIES OF In_2O_3 TRANSISTORS

Temperature	μ_{FE} ($cm^2/V{\cdot}s$)	V_{th} (V)	SS (mV/dec)
25°C	84	-0.84	73.3
45°C	82	-0.88	70
65°C	81.3	-0.9	69.7
85°C	80	-0.94	69.3

at 85 °C using the transmission-line (TLM) model. The R_c of the In_2O_3 device is 257 $\Omega \cdot \mu$m at a carrier density (n_s) of $1.9 \times 10^{13} cm^{-2}$ as shown in Fig. 3(f). Table I summarizes the electrical properties of In_2O_3 transistors at different temperatures.

To further evaluate the interface quality of the In_2O_3 transistor and its impact on high-temperature carrier transport, the low-frequency noise behaviors of the In_2O_3 devices were measured at $V_{ds} = 0.05$ V. Fig. 4(a) and (b) show the normalized current spectral density S_{id}/I_d^2 with five different back-gate biases in the 1-1000 Hz range at 25 and 85 °C, respectively. It nearly follows a $1/f$ rule from 25 to 85 °C at different gate voltages, which indicates that the flicker noise ($1/f$ noise) is the dominant component of the LFN in the In_2O_3 transistor. The S_{id}/I_d^2 values extracted at frequencies of 100 Hz and $(g_m/I_d)^2$ as a function of drain current I_d are shown in Fig. 4(c) at 25 °C and 4(d) at 85 °C, respectively. With a drain current below 10 μA, the S_{id}/I_d^2 curve with a slope of ~ -2 shows good proportionality with $(g_m/I_d)^2$, indicating that a carrier number fluctuation [21], [22], caused by the trapping and release of carriers in the gate stack, dominates the $1/f$ noise. There are slope deviations observed between the S_{id}/I_d^2 and $(g_m/I_d)^2$ curves in the range of drain current over 10 μA, in agreement with the correlated mobility fluctuation model which means that the mobility scattering is more significant [21]. The extracted curves at 25 °C and 85 °C show negligible differences, verifying the absence of interface traps during thermal activation. The average oxide trap density N_{ot} of the In_2O_3 device at 85 °C is $\sim 3.4 \times 10^{19}$ $eV^{-1}cm^{-3}$ according to the input-referred voltage noise spectral density S_{vg} at the threshold voltage [23], [24].

Fig. 5(c) and (d) show the time evolution of the ΔV_{th} of In_2O_3 FETs with $L_{ch} = 2$ μm under NBS and PBS at 25 and 85 °C, respectively. The magnitude of the V_{th} shift increases at 85 °C due mainly to faster generation of acceptor-like (for NBS) or donor-like traps (for PBS) at high temperatures [25], [26]. Overall, the V_{th} shifts of NBS and PBS are still relatively small compared with those reported in a previous study on In_2O_3 devices [8], indicating the effective passivation of the In_2O_3 channel by O_3-based ALD HfO_2.

To investigate the possible contribution of the contacts to the LFN, In_2O_3 devices with different channel lengths were measured with V_{bg} from the threshold voltage region to the accumulation region. Fig. 5 shows the dependence of S_{id}/I_d^2 on the channel length of the In_2O_3 devices at 85 °C. With increasing gate voltage to the accumulation region ($V_{bg} = 2.8$ V), the noise level decreases over an order of magnitude and strongly depends on the channel length. The obtained slope of the region in the log-log plot is ~ -1, which indicates that the LFN is generated mainly in the channel region of the In_2O_3 transistors. To benchmark the performance of the In_2O_3 device in this work, we compare the drain current noise level S_{id}/I_d^2 against the device channel area $W_{ch} \times L_{ch}$ with that of typical oxide FETs in the literature, as shown in Fig. 5(b) [22], [27], [28], [29], [30], [31]. The In_2O_3 device used in this work showed a significant improvement in the noise level, which decreased to 1.4×10^{-11} $\mu m^2 Hz^{-1}$ at 100 Hz due to the high carrier mobility and on-state current.

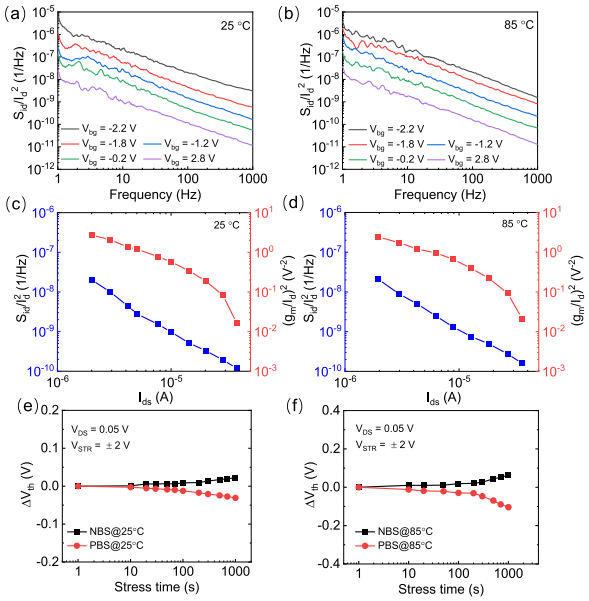


Fig. 4. Normalized current noise spectral density S_{id}/I_d^2 of the passivated In_2O_3 FET with $L_{ch} = 0.2$ μm at five different back-gate biases for (a) 25 °C, and (b) 85 °C. Normalized current noise spectral density S_{id}/I_d^2 and $(g_m/I_d)^2$ as a function of drain current I_d (c) at 25 °C and (d) at 85 °C. Positive bias stress (PBS) and negative bias stress (NBS) at (e) 25 °C and (f) 85 °C.

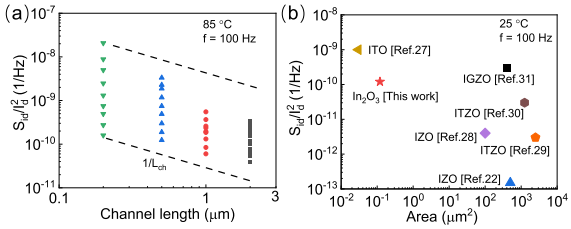


Fig. 5. (a) Dependence of S_{id}/I_d^2 on the channel length of devices at 85 °C. The data points displayed at a fixed channel length (in the same color) correspond to different gate voltages. (b) Benchmark of the current noise spectral density S_{id}/I_d^2 versus the channel area for different oxide transistors.

IV. CONCLUSION

In conclusion, we report high-performance back-gate In_2O_3 transistors with a high field-effect mobility of 80 $cm^2/V{\cdot}s$ and a high on/off ratio of $\sim 10^{10}$ at 85 °C. An ultralow average rate of voltage shift of -0.8 mV/K and excellent bias-thermal stability are achieved. The low-frequency noise level of In_2O_3 transistors with $L_{ch} = 0.2$ μm at different temperatures remains almost the same, excluding the effect of the interface trap during thermal activation. Compared with those of the other process methods, the excellent temperature dependence of the In_2O_3 transistor shows the effective passivation of oxygen vacancies by O_3-based ALD HfO_2, indicating promising applications in monolithic three-dimensional integration.

Authorized licensed use limited to: Huazhong University of Science and Technology. Downloaded on February 04,2025 at 06:05:16 UTC from IEEE Xplore. Restrictions apply.

REFERENCES

[1] C. Fenouillet-Beranger, L. Brunet, P. Batude, L. Brevard, X. Garros, M. Cassé, J. Lacord, B. Sklenard, P. Acosta-Alba, S. Kerdilès, A. Tavernier, C. Vizioz, P. Besson, R. Gassilloud, J.-M. Pedini, J. Kanyandekwe, F. Mazen, A. Magalhaes-Lucas, C. Cavalcante, D. Bosch, M. Ribotta, V. Lapras, M. Vinet, F. Andrieu, and J. Arcamone, "A review of low temperature process modules leading up to the first (=500 °C) planar FDSOI CMOS devices for 3-D sequential integration," *IEEE Trans. Electron Devices*, vol. 68, no. 7, pp. 3142–3148, Jul. 2021, doi: 10.1109/TED.2021.3084916.

[2] M. Si, A. Murray, Z. Lin, J. Andler, J. Li, J. Noh, S. Alajlouni, C. Niu, X. Lyu, D. Zheng, K. Maize, A. Shakouri, S. Datta, R. Agrawal, and P. D. Ye, "BEOL compatible indium-tin-oxide transistors: Switching of ultrahigh-density 2-D electron gas over 0.8 × 10^{14}/cm^2 at oxide/oxide interface by the change of ferroelectric polarization," *IEEE Trans. Electron Devices*, vol. 68, no. 7, pp. 3195–3199, Jul. 2021, doi: 10.1109/TED.2021.3061038.

[3] S. Li, M. Tian, C. Gu, R. Wang, M. Wang, X. Xiong, X. Li, R. Huang, and Y. Wu, "BEOL compatible 15-nm channel length ultrathin indium-tin-oxide transistors with ion = 970 μA/μm and on/off ratio near 10^{11} at V_{ds} = 0.5 V," in *IEDM Tech. Dig.*, Dec. 2019, pp. 3.5.1–3.5.4, doi: 10.1109/IEDM19573.2019.8993488.

[4] M. Si, Z. Lin, Z. Chen, X. Sun, H. Wang, and P. D. Ye, "Scaled indium oxide transistors fabricated using atomic layer deposition," *Nature Electron.*, vol. 5, no. 3, pp. 164–170, Feb. 2022, doi: 10.1038/s41928-022-00718-w.

[5] S. Subhechha, N. Rassoul, A. Belmonte, R. Delhougne, K. Banerjee, G. L. Donadio, H. Dekkers, M. J. van Setten, H. Puliyalil, M. Mao, S. Kundu, M. Pak, L. Teugels, D. Tsvetanova, N. Bazzazian, L. Klijs, H. Hody, A. Chasin, J. Heijlen, L. Goux, and G. S. Kar, "First demonstration of sub-12 nm L_g gate last IGZO-TFTs with oxygen tunnel architecture for front gate devices," in *Proc. Symp. VLSI Technol.*, Jun. 2021, pp. 1–2.

[6] S. Li, M. Tian, Q. Gao, M. Wang, T. Li, Q. Hu, X. Li, and Y. Wu, "Nanometre-thin indium tin oxide for advanced high-performance electronics," *Nature Mater.*, vol. 18, no. 10, pp. 1091–1097, Oct. 2019, doi: 10.1038/s41563-019-0455-8.

[7] M. Si, J. Andler, X. Lyu, C. Niu, S. Datta, R. Agrawal, and P. D. Ye, "Indium–tin-oxide transistors with one nanometer thick channel and ferroelectric gating," *ACS Nano*, vol. 14, no. 9, pp. 11542–11547, Sep. 2020, doi: 10.1021/acsnano.0c03978.

[8] Z. Zhang, Z. Lin, C. Niu, M. Si, M. A. Alam, and P. D. Ye, "Ultra-high bias stability of ALD In_2O_3 FETs enabled by high temperature O_2 annealing," in *Proc. IEEE Symp. VLSI Technol. Circuits (VLSI Technol. Circuits)*, Jun. 2023, pp. 1–2, doi: 10.23919/vlsitechnologyandcir57934.2023.10185292.

[9] K. Heo, K.-S. Cho, J. Y. Choi, S. Han, Y. S. Yu, Y. Park, G. Yoo, J.-H. Park, S. W. Hwang, and S. Y. Lee, "Temperature-dependent electrical characterization of amorphous indium zinc oxide thin-film transistors," *IEEE Trans. Electron Devices*, vol. 64, no. 8, pp. 3183–3188, Aug. 2017, doi: 10.1109/TED.2017.2717935.

[10] M. D. Hossain Chowdhury, P. Migliorato, and J. Jang, "Low temperature characteristics in amorphous indium-gallium-zinc-oxide thin-film transistors down to 10 k," *Appl. Phys. Lett.*, vol. 103, no. 15, Oct. 2013, Art. no. 152103, doi: 10.1063/1.4824875.

[11] H. Godo, D. Kawae, S. Yoshitomi, T. Sasaki, S. Ito, H. Ohara, H. Kishida, M. Takahashi, A. Miyanaga, and S. Yamazaki, "Temperature dependence of transistor characteristics and electronic structure for amorphous In–Ga–Zn-Oxide thin film transistor," *Jpn. J. Appl. Phys.*, vol. 49, no. 3S, Mar. 2010, Art. no. 03CB04, doi: 10.1143/jjap.49.03cb04.

[12] J. Jeong, G. Jun Lee, J. Kim, S. Moon Jeong, and J.-H. Kim, "Analysis of temperature-dependent electrical characteristics in amorphous In-Ga-Zn-O thin-film transistors using gated-four-probe measurements," *J. Appl. Phys.*, vol. 114, no. 9, Sep. 2013, Art. no. 094502, doi: 10.1063/1.4819886.

[13] K. Hoshino and J. F. Wager, "Operating temperature trends in amorphous In–Ga–Zn–O thin-film transistors," *IEEE Electron Device Lett.*, vol. 31, no. 8, pp. 818–820, Aug. 2010, doi: 10.1109/LED.2010.2049980.

[14] H. Y. Lee, J. S. Hur, I. Cho, C. H. Choi, S. H. Yoon, Y. Kwon, B. Shong, and J. K. Jeong, "Comparative study on indium precursors for plasma-enhanced atomic layer deposition of In_2O_3 and application to high-performance field-effect transistors," *ACS Appl. Mater. Interface*, vol. 15, no. 44, pp. 51399–51410, Oct. 2023, doi: 10.1021/acsami.3c11796.

[15] A. Charnas, Z. Lin, Z. Zhang, and P. D. Ye, "Atomically thin In_2O_3 field-effect transistors with 10^{17} current on/off ratio," *Appl. Phys. Lett.*, vol. 119, no. 26, Dec. 2021, Art. no. 263503, doi: 10.1063/5.0075166.

[16] S. Wahid, A. Daus, J. Kwon, S. Qin, J.-S. Ko, H.-S. Philip Wong, and E. Pop, "Effect of top-gate dielectric deposition on the performance of indium tin oxide transistors," *IEEE Electron Device Lett.*, vol. 44, no. 6, pp. 951–954, Jun. 2023, doi: 10.1109/LED.2023.3265316.

[17] J. Zhang, Z. Lin, Z. Zhang, K. Xu, H. Dou, B. Yang, A. Charnas, D. Zheng, X. Zhang, H. Wang, and P. D. Ye, "Back-end-of-line-compatible scaled InGaZnO transistors by atomic layer deposition," *IEEE Trans. Electron Devices*, vol. 70, no. 12, pp. 6651–6657, Dec. 2023, doi: 10.1109/TED.2023.3312357.

[18] C. Gu, Q. Hu, S. Zhu, Q. Li, M. Zeng, H. Liu, J. Kang, S. Liu, and Y. Wu, "High-performance short-channel top-gate indium-tin-oxide transistors by optimized gate dielectric," *IEEE Electron Device Lett.*, vol. 44, no. 5, pp. 837–840, May 2023, doi: 10.1109/LED.2023.3262684.

[19] M. J. Kim, S. Lee, E. H. Kim, J. H. Lim, and J. K. Jeong, "Theoretical modeling of a temperature-dependent threshold-voltage shift in self-aligned coplanar IZTO thin-film transistors," *ACS Appl. Electron. Mater.*, vol. 5, no. 6, pp. 3010–3022, May 2023, doi: 10.1021/acsaelm.2c01754.

[20] C. R. Allemang and R. L. Peterson, "Passivation of thin channel zinc tin oxide TFTs using Al_2O_3 deposited by O_3-based atomic layer deposition," *IEEE Electron Device Lett.*, vol. 40, no. 7, pp. 1120–1123, Jul. 2019, doi: 10.1109/LED.2019.2914238.

[21] C. Wei, Y. Jiang, Y.-Z. Xiong, X. Zhou, N. Singh, S. C. Rustagi, G. Qiang Lo, and D.-L. Kwong, "Impact of gate electrodes on $1/f$ noise of gate-all-around silicon nanowire transistors," *IEEE Electron Device Lett.*, vol. 30, no. 10, pp. 1081–1083, Oct. 2009, doi: 10.1109/LED.2009.2027614.

[22] Y. Liu, H. He, Y.-Y. Chen, R. Chen, L. Wang, S. Cai, and X. Xiong, "Temperature-dependent low-frequency noise in indium–zinc–oxide thin-film transistors down to 10 k," *IEEE Trans. Electron Devices*, vol. 66, no. 5, pp. 2192–2197, May 2019, doi: 10.1109/TED.2019.2902449.

[23] M. Scarpino, S. Gupta, D. Lin, A. Alian, F. Crupi, N. Collaert, A. Thean, and E. Simoen, "Border traps in InGaAs nMOSFETs assessed by low-frequency noise," *IEEE Electron Device Lett.*, vol. 35, no. 7, pp. 720–722, Jul. 2014, doi: 10.1109/LED.2014.2322388.

[24] B. C. Wang, S. L. Wu, Y. Y. Lu, S. J. Chang, J. F. Chen, S. C. Tsai, C. H. Hsu, C. W. Yang, C. G. Chen, O. Cheng, and P. C. Huang, "Comparison of the trap behavior between ZrO_2 and HfO_2 gate stack nMOSFETs by 1/f noise and random telegraph noise," *IEEE Electron Device Lett.*, vol. 34, no. 2, pp. 151–153, Feb. 2013, doi: 10.1109/LED.2012.2226698.

[25] Y.-P. Chen, M. Si, B. K. Mahajan, Z. Lin, P. D. Ye, and M. A. Alam, "Positive bias temperature instability and hot carrier degradation of back-end-of-line, nm-thick, In_2O_3 thin-film transistors," *IEEE Electron Device Lett.*, vol. 43, no. 2, pp. 232–235, Feb. 2022, doi: 10.1109/LED.2021.3134902.

[26] A. Charnas, M. Si, Z. Lin, and P. D. Ye, "Improved stability with atomic-layer-deposited encapsulation on atomic-layer In_2O_3 transistors by reliability characterization," *IEEE Trans. Electron Devices*, vol. 69, no. 10, pp. 5549–5555, Oct. 2022, doi: 10.1109/TED.2022.3198926.

[27] C. Gu, Q. Hu, Q. Li, S. Zhu, J. Kang, and Y. Wu, "1/f noise of short-channel indium tin oxide transistors under stress," *Appl. Phys. Lett.*, vol. 122, no. 25, Jun. 2023, Art. no. 252104, doi: 10.1063/5.0147577.

[28] W. Ye, Y. Liu, B. Wang, J. Huang, X. Xiong, and W. Deng, "Low-frequency noise modeling of amorphous indium–zinc-oxide thin-film transistors," *IEEE Trans. Electron Devices*, vol. 69, no. 11, pp. 6154–6159, Nov. 2022, doi: 10.1109/TED.2022.3206274.

[29] Y. Chen, Y. Liu, S. Deng, R. Chen, J. Zhang, H.-S. Kwok, and W. Zhong, "Low-frequency noise in InSnZnO thin film transistors with high-quality SiO_2 gate oxide stacks," *Appl. Phys. Lett.*, vol. 124, no. 2, Jan. 2024, Art. no. 023501, doi: 10.1063/5.0174250.

[30] Z. Jiang, M. Zhang, S. Deng, M. Wong, and H.-S. Kwok, "Degradation of InSnZnO thin-film transistors under negative bias stress," *IEEE Trans. Electron Devices*, vol. 70, no. 12, pp. 6381–6386, Dec. 2023, doi: 10.1109/TED.2023.3327975.

[31] W. Shin, E. C. Park, R.-H. Koo, D. Kwon, D. Kwon, and J.-H. Lee, "Low-frequency noise characteristics of indium–gallium–zinc oxide ferroelectric thin-film transistors with metal–ferroelectric–metal–insulator–semiconductor structure," *Appl. Phys. Lett.*, vol. 122, no. 15, Apr. 2023, Art. no. 152905, doi: 10.1063/5.0140953.

Authorized licensed use limited to: Huazhong University of Science and Technology. Downloaded on February 04,2025 at 06:05:16 UTC from IEEE Xplore. Restrictions apply.